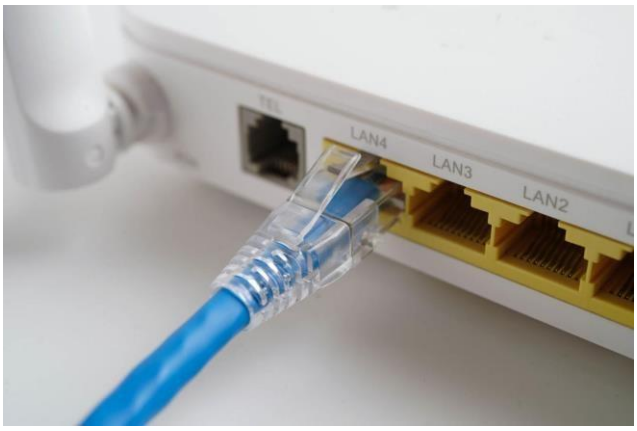
Assignment-4

Day 4: Network Cable Types and Tools Theory Types of cables (Ethernet, fiber optics) and tools (RJ45 crimper). Practical Create a straight-through Ethernet cable

Task-1: List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

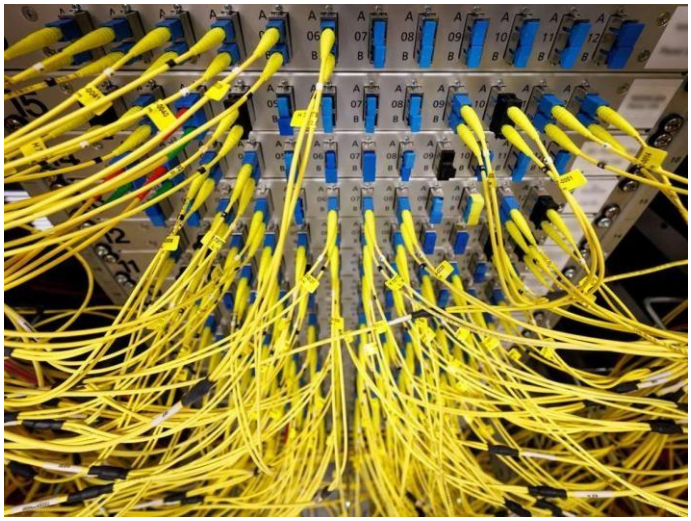
Feature	Ethernet Cable	Fiber Optic Cable
1. Transmission medium	Uses electrical signals through copper wires.	Uses light signals through optical fibers.
2. Speed and distance	Commonly used for connections up to 100 meters per copper Ethernet link; speed depends on the cable category and equipment.	Supports very high speeds over much longer distances, depending on the fiber type and equipment.
3. Cost and interference	Generally cheaper and easier to install, but can be affected by electromagnetic interference.	Usually more expensive to install and requires compatible optical equipment, but is resistant to electromagnetic interference.

Real-world examples



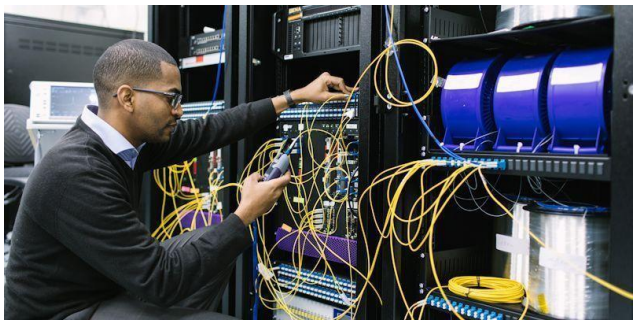
1. Ethernet cable — Home or office network

Connects a computer, gaming PC, or smart TV to a router or network switch for a stable wired connection.



2. Fiber optic cable — Data center

Connects network switches and servers over high-speed links, especially where high bandwidth is required.

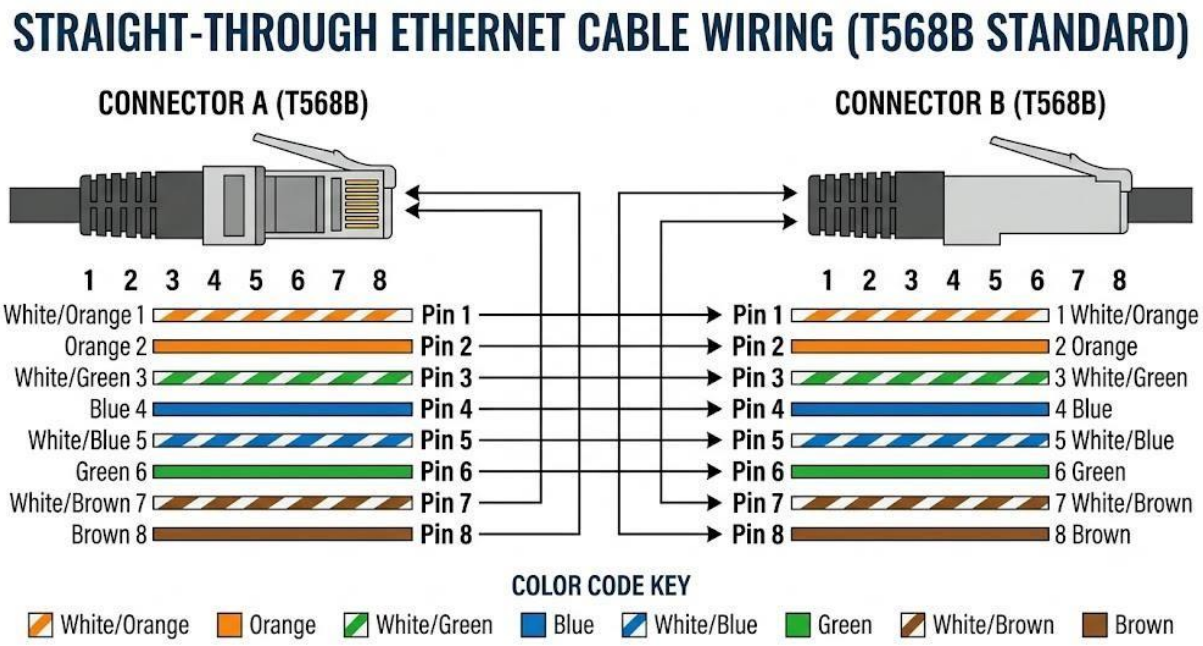


3. Fiber optic cable — College campus backbone

Connects network rooms in different buildings over long distances with high-speed data transmission.

Conclusion: Ethernet cables are cost-effective for short-distance device connections, while fiber optic cables are ideal for high-speed, long-distance network connections.

Task-2: Draw and label a diagram showing the internal wiring of a straightthrough Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



Task-3: Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Step-by-Step Guide to Attaching an RJ45 Connector to a CAT6 Cable

Aim

To learn how to attach an RJ45 connector to a CAT6 Ethernet cable using an RJ45 crimping tool.

Tools and Materials Required

1. CAT6 Ethernet cable

2. RJ45 connector (8P8C)
3. RJ45 crimping tool
4. Wire stripper or cable cutter
5. Ethernet cable tester

Procedure

Step 1: Strip the Outer Jacket

Cut the end of the CAT6 cable neatly. Use a wire stripper to remove approximately 25–30 mm of the outer jacket without damaging the insulated wires inside.

Step 2: Separate the Wires

Inside the cable, there are four twisted pairs of wires, making a total of eight individual wires. Carefully separate the pairs and straighten the wires gently.

Step 3: Arrange the Wires in the Correct Order

Arrange the eight wires according to the T568B wiring standard in the following order:

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

Ensure that the wires are arranged correctly before proceeding.

Step 4: Trim the Wires

Hold all eight wires together in the correct order and cut their ends evenly using a wire cutter. Make sure the wires are short enough to fit properly into the RJ45 connector but long enough to reach its front contacts.

Step 5: Insert the Wires into the RJ45 Connector

Carefully insert the eight wires into the RJ45 connector. Ensure that each wire enters its separate channel and reaches the front of the connector. The outer jacket of the cable should also extend inside the connector so that it can be secured properly.

Step 6: Crimp the Connector

Place the RJ45 connector into the appropriate slot of the crimping tool. Squeeze the handles firmly to press the metal contacts into the wires and secure the cable jacket. Remove the connector and check that it is firmly attached.

Step 7: Test the Cable

Connect both ends of the cable to an Ethernet cable tester. Check that all eight wires are connected in the correct sequence. If the tester indicates a fault, inspect the connections and repeat the termination if necessary.

Precautions

1. Do not damage the insulated wires while stripping the outer jacket.
2. Maintain the correct T568B wire sequence.
3. Avoid excessive untwisting of the wire pairs.
4. Ensure all eight wires reach the front of the connector.
5. Use a CAT6-compatible RJ45 connector.
6. Crimp the connector firmly and test the cable before use.

Result

The RJ45 connector was attached to the CAT6 Ethernet cable using a crimping tool. The cable was tested to verify the correct connection of all eight conductors.

TAsk-4: Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

For setting up a gaming café with 10 PCs, I would choose Ethernet cables (CAT6) and an RJ45 crimping tool. Ethernet cables provide fast, reliable, and low-latency connections that are suitable for LAN gaming. The RJ45 crimper is used to attach RJ45 connectors to the cables, allowing the PCs to connect to a network switch. This setup is cost-effective, easy to install, and ideal for a small gaming café.